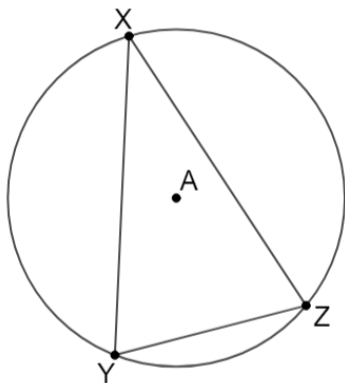


Geometry
Unit 11
Lesson 3 Practice

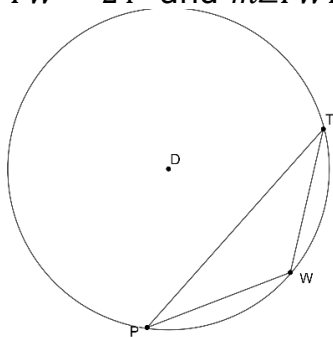
Name **Answer Key**
Date _____
Period _____

Circle A is shown, where points X, Y , and Z are on Circle A and create $\triangle XYZ$. $m\widehat{XZ} = 140^\circ$, $m\angle ZXY = 30^\circ$, and $m\angle XYZ = (2x - 18)^\circ$.



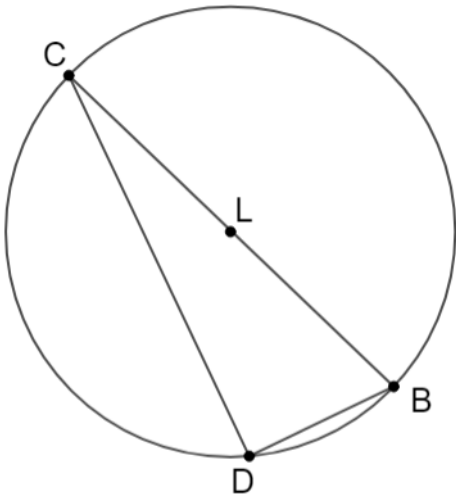
- Find the value of x . $2x - 18 = \frac{140}{2}$
 $2x = 88$
 $x = 44$
- Find the $m\angle XYZ$. $m\angle XYZ = 2(44) - 18 = 70^\circ$
- Find the $m\widehat{YZ}$. $2(30^\circ) = 60^\circ = m\widehat{YZ}$.
- Find $m\angle YZX$. $m\angle YZX + 30 + 70 = 180$
 $m\angle YZX = 80^\circ$

Circle D is shown, where points P, T , and W are on Circle D and create $\triangle PTW$. $m\angle PTW = 24^\circ$ and $m\angle TWP = 140^\circ$.



- Find $m\widehat{PT}$. $m\widehat{PT} = 2(140) = 280^\circ$
- Find $m\widehat{TW}$. $m\angle P = 180 - 140 - 24 = 16^\circ$; $m\widehat{TW} = 2(16^\circ) = 32^\circ$

Circle L has points B, C , and D on Circle L and diameter $\overline{BC}$. $\triangle BCD$ is inscribed in the circle. $m\widehat{CD} = 160^\circ$, $m\angle DCB = 10^\circ$, and $m\angle DBC = (2x + 20)^\circ$.



7. What is the $m\angle CDB$? $m\angle CDB = \frac{180}{2} = 90^\circ$
8. What is the value of x ? $2x + 20 = \frac{160}{2}$; $2x = 60$; $x = 30$
9. Determine $m\angle DBC$. $2x + 20 = 2(30) + 20 = 80 = m\angle DBC$
10. Determine $m\widehat{DBC}$. $m(\widehat{DBC}) = 360 - 160 = 200^\circ$